Pablo Piantanida

Academic Curriculum Vitae

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Executive Profile

Full Professor and research leader developing the **mathematical and information-theoretic foundations of reliable AI**. My research studies what information learning systems preserve, what decisions it supports, and how reliability can be assessed under limited evidence, computation, and communication. It connects information theory, statistics, and learning theory with modern machine learning to produce fundamental results and experimentally validated methods.

Since 2022, I have served as **Director of the CNRS International Research Laboratory on Learning Systems (ILLS)** in Montréal, where I shape the scientific strategy, develop institutional partnerships, and coordinate France–Canada research and training initiatives connecting foundational AI with language, healthcare, scientific discovery, and engineering.

My current research centres on **evaluation, uncertainty, and failure prediction for foundation models**, extending these methods to adaptive evaluation and agentic systems. I develop statistically principled approaches to error detection, distribution shift, uncertainty quantification, and privacy and generative-system auditing. The goal is decision-aware evaluation: selecting informative tests, determining when evidence is sufficient, and supporting decisions to deploy, abstain, acquire further information, or request human oversight under explicit risk requirements.

My emerging research programme develops **resource-aware agentic and collaborative intelligence**. Building on interactive source coding, the collaborative information bottleneck, and pragmatic communication, it studies how AI systems should acquire evidence, allocate computation, use tools, and coordinate with humans and other agents. The objective is to derive adaptive policies with measurable trade-offs among task performance, reliability, communication, privacy, and human oversight.

Research Impact at a Glance

Publications	**249 journal and refereed conference publications**, including 60 journal articles and 189 international conference papers.
Research funding	EUR 2.31M + CAD 504K + USD 60K in competitive funding secured as PI, co-PI, co-holder, or chair holder, plus EUR 120K in in-kind GPU resources.
Industry	Research collaborations with **GE HealthCare, Mistral AI, Safran, IBM, Ekimetrics, and Air Liquide** on medical imaging, AI-assisted control, tabular foundation models, and generative AI for molecular generation.
Doctoral supervision	**22 completed Ph.D. students** and **12 current Ph.D. students**, supervised or co-supervised.
Profiles	Google Scholar | ORCID: 0000-0002-8717-2117
Impact	h-index: 36; 5,800+ citations (Google Scholar).

Research Trajectory and Programme

Unifying scientific question

My research is organised around one question: **how can AI systems use information to make reliable decisions when evidence, computation, and communication are limited?** This question connects my foundations in information theory with research on representations, evaluation, uncertainty, and collaboration. The central objective is to determine what the available information supports, when further evidence is necessary, and how systems should acquire or exchange it.

The programme combines **fundamental limits and mathematical characterisations**, **actionable methods and evaluation protocols**, and **credible empirical validation**. Projects are selected for their ability to connect a precise mathematical question with a consequential decision and a measurable practical outcome.

From information theory to the foundations of AI

My foundational research studied communication and distributed decision-making under channel uncertainty, interference, secrecy, privacy, and resource constraints. It developed achievability and converse techniques for multiterminal systems, including *Distributed Information-Theoretic Clustering* (Information and Inference, 2022) and *Three-Terminal Interactive Lossy Source Coding* (IEEE Trans. Inf. Theory, 2017). These contributions ask what tasks can be accomplished from partial observations and limited communication, and how much information is fundamentally necessary.

This perspective extends naturally to learning systems. Information theory and statistical decision theory provide operational notions of relevance, sufficiency, deficiency, distinguishability, leakage, and value of information. My work on representations, task inclusion, privacy, and prediction-error detection connects these notions to statistical inference and modern machine learning. Interactive coding, the collaborative information bottleneck, and pragmatic communication provide foundations for the emerging programme on collaborative AI.

Research programme for 2027–2031

Axis I – Information-theoretic foundations of representations and learning. Develop mathematical principles for learning representations that preserve task-relevant information, generalise from limited data, and control sensitive-information leakage. Building on my contributions to information-theoretic representation learning, information estimation, few-shot learning, and multi-teacher distillation, this axis will connect representation design with statistical guarantees and downstream performance. Information sufficiency and statistical deficiency will provide frameworks for comparing representations, tasks, and learning algorithms through their consequences for decisions. Expected contributions include estimators, learning objectives, and finite-data bounds for representation quality, transfer, and privacy, with selected extensions to machine unlearning.

Axis II – Uncertainty, failure prediction, and adaptive evaluation. Develop statistically principled methods to assess when predictions, generated outputs, and multi-step behaviours are unreliable. Building on my work on misclassification and regression-error detection, information-geometric OOD detection, language-model evaluation, and generative-system auditing, this axis will connect uncertainty indicators with task-specific risks and decisions. A central extension is adaptive evaluation: selecting informative tests and determining when evidence is sufficient despite uncertain calibration, imperfect automated judges, and limited human feedback. The programme will de-

rive test-selection and stopping procedures, characterise their evidence requirements, and establish guarantees under explicit assumptions on sampling and distribution shift. Validation will measure evaluation cost, missed failures, false alarms, and the reliability of model-comparison decisions.

Axis III – Distributed inference and resource-aware collaborative AI. Develop mathematical and algorithmic foundations for systems that exchange information and coordinate decisions under uncertainty. This axis builds on my work on interactive source coding, collaborative hypothesis testing, distributed information bottlenecks, and pragmatic multi-turn communication. It will study which information agents should communicate, how they should update beliefs from partial observations and dialogue, and when interaction improves decisions enough to justify its cost. The programme will extend these foundations to policies that choose among further computation, information acquisition, tool use, communication, and human assistance. Initial studies will consider cooperative agents with private information and imperfect models of one another, quantifying trade-offs among decision quality, communication, privacy, and resource use. Comparisons with strong single-agent alternatives under matched budgets will establish when collaboration provides a measurable advantage.

Integration across the axes. Axis I studies how useful information is learned and represented. Axis II determines what evidence supports an assessment of system reliability. Axis III studies how systems should acquire and exchange information to improve subsequent decisions. Information theory and statistical decision theory connect these questions through common notions of relevance, uncertainty, risk, and resource constraints.

Application and validation strategy. Language and knowledge-intensive tasks will provide a shared environment for representation learning, evaluation, and collaborative decision-making. One or two flagship projects will extend the programme to molecular discovery, medical imaging, or aircraft assistance, according to partner access, data availability, and credible validation conditions. Each project will connect a mathematical question with an actionable method and a measurable outcome: improved transfer with limited labelled data; lower evaluation cost at a controlled decision-error rate; better failure detection at a fixed false-alarm rate; or improved task performance with fewer resources or human interventions. Controlled experiments will test theoretical predictions, followed by validation on realistic tasks with domain partners.

Selected Contributions and Directions

Information-theoretic foundations of representations and learning

- Representation learning and generalisation: *The Role of Mutual Information in Variational Classifiers* (Machine Learning, 2023), *Information Maximization for Few-Shot Learning* (NeurIPS 2020), and *Task-Agnostic Multi-Teacher Distillation* (NeurIPS 2025).
- Information estimation and privacy: *Differential Entropy Estimation* (ICML 2022), *Mutual-Information Estimation for Disentanglement* (ACL 2021), g*-Leakage Estimation* (CCS 2020), and *Fundamental Limits of Membership Inference* (JMLR 2025).
- Statistical comparison of representations and tasks: *When is an Embedding Model More Promising than Another?* (NeurIPS 2024) and *Statistical Deficiency for Task Inclusion Estimation* (ACL 2025), providing foundations for model selection and transfer.

Uncertainty, failure prediction, and adaptive evaluation

- Robustness, distribution shift, and prediction-error detection: *Fisher–Rao Regularization* (IEEE TPAMI 2023), *DOCTOR* (NeurIPS 2021), *IGEOOD* (ICLR 2022), *Relative Uncertainty* (ICLR

2024), and *Beyond the Norms* (ICML 2024).

- Evaluation of generated language and generative-system auditing: *InfoLM* (AAAI 2022 Outstanding Student Paper), *COSMIC* (ACL 2024), *Unsupervised Hallucination Detection*, and *MOSAIC* (Findings of ACL 2025).
- The *REGARD project* addresses real-world evaluation of generative AI. An emerging direction connects uncertainty and failure detection with adaptive test selection and stopping under limited evidence, uncertain calibration, and imperfect feedback.

Distributed inference and resource-aware collaborative AI

- Fundamental limits of distributed inference and interaction: *Collaborative Distributed Hypothesis Testing with General Hypotheses* (ISIT 2016), *The Three-Terminal Interactive Lossy Source Coding Problem* (IEEE Trans. Inf. Theory, 2017), and *Collaborative Information Bottleneck* (IEEE Trans. Inf. Theory, 2019).
- Pragmatic communication and multi-turn interaction: *Rational Retrieval Acts* (SIGIR 2025) and *Collaborative Rational Speech Act* (EMNLP 2025), connecting information-theoretic principles with retrieval and collaborative dialogue.
- Emerging directions study the value of evidence acquisition, computation, communication, and human assistance, and establish when collaboration improves decisions under matched resource budgets.

Current and Previous Appointments

January 2022 – Present **Director, CNRS International Research Laboratory on Learning Systems (ILLS, IRL2020)**, *International Research Laboratory jointly supported by McGill University, ÉTS, Mila, Université Paris-Saclay, and CentraleSupélec*, Montréal, QC

Sep. 2023 – Present **Associate Academic Member**, *Mila - Quebec AI Institute*, Montréal, QC, Canada

Sep. 2022 – Present **Adjunct Professor**, *École de technologie supérieure (ÉTS)*, Montréal, QC, Canada

Sep. 2020 – Present **Full Professor**, *CentraleSupélec - Université Paris-Saclay*, France

Sep. 2019 – 2022 **Associate Research Memeber**, *Comète Inria Team (Laboratoire d'informatique de l'École polytechnique)*, École polytechnique, Palaiseau, France

Sep. 2019 – 2020 **Visiting Researcher**, *Laboratoire de Mathématiques d'Orsay (LMO)*, Orsay, France

June 2018 – 2019 **Visiting Researcher**, *Université de Montréal (UdeM), Mila - Quebec AI Institute*, Montréal, QC, Canada

Sep. 2015 – 2020 **Associate Professor**, *École CentraleSupélec, Laboratoire des Signaux et Systèmes (L2S, UMR 8506), CNRS, Université Paris Saclay, France*

Sep. 2007 – 2015 **Assistant Professor**, *École Supélec, Département de Télécommunications, France*

Research and Institutional Leadership

2022 – Present **Director and founding scientific lead**, *CNRS International Research Laboratory on Learning Systems (ILLS, IRL2020)*, Montréal, Canada, Joint laboratory involving McGill University, ÉTS, Mila, CNRS, Université Paris-Saclay, and CentraleSupélec
Lead the laboratory's international research strategy, institutional partnerships, scientific programme, and France–Canada research and training initiatives

2017 – 2020 **Scientific Coordinator of the IRP (International Research Project) "Information, Learning and Control"**, *France–Canada collaborative research network*, 8 Canadian and French institutions: McGill University, Université de Montréal, ÉTS, Concordia University, INRS, Université Paris-Sud, CNRS, and CentraleSupélec, Coordination of an international research network in information, learning, and control, laying the groundwork for long-term transatlantic collaboration

2017 – 2018 **Curriculum Coordinator, ST4 track, SCOC program**, *CentraleSupélec*, Led first-year course and curriculum design within the engineering program

2018 – 2019 **Co-coordinator of the International Summer School on Artificial Intelligence**, *Direction Internationale, CentraleSupélec*

2015 – 2019 **Coordinator, research group ITA**, *Information Theory and its Applications (7 staff members out of 15)*, Laboratoire des Signaux et Systèmes, CNRS-UMR8506

2015 – 2019 **Co-responsible for CentraleSupélec of Master**, *Multimedia Networking*, E3A / Electrical Engineering, Université Paris Saclay

Competitive Funding and Research Support (selected)

2025 – 2029 **ANR–Canada Tri-agency joint call on AI (ANR; NSERC/SSHRC/CIHR) with IVADO support**, *PI*, "REGARD — Robust Evaluation of Generative AI in Real-World Environments", 250K Euros
Collaboration between University of Waterloo, McGill University, Aix-Marseille Université, CNRS, CentraleSupélec, and Orange Labs.

2026 - 2030 **Holder of the AI Cluster DATAIA Paris-Saclay - Chaires modulaires**, *Mathematical Principles of Learning and Communication in Attention-Based Models*, Jointly with Prof. Claire Boyer at Laboratoire des Mathématiques d'Orsay (LMO, UPSaclay), 500K Euros

2025 – 2027 **IVADO R2 Group**, *co-PI*, "Cognitive Communications for Multi-Agent Reinforcement Learning", 50K CAD
Collaboration between McGill University, ETS, Université Laval

2026 – 2029 **CIFRE - GE HealthCare (industry collaboration)**, *PI*, "Foundation Models for 3D Image Segmentation", 90K Euros
(excluding salary and other expenses).

2026 – 2029 **CIFRE - Ekimetrics (industry collaboration)**, *PI*, "Foundation Models for Tabular Data", 90K Euros
(excluding salary and other expenses).

2025 – 2029 **Mitacs - Safran (industry collaboration)**, *PI*, "AI-based Control Method for Aircraft Flight Assistance", 160K CAD
Collaboration between ETS and Safran.

2025 – 2026 **Mistral AI (industry collaboration)**, *PI*, "MolRGen: A Training and Evaluation Setting for De Novo Molecular Generation with Reasoning Models", 120K Euros (GPU computing resources).

2024 – 2027 **CIFRE - Air Liquide (industry collaboration)**, *PI*, "Generative AI for Industrial Applications", 75K Euros (excluding salary and other expenses).

2024 – 2026 **FRQS - HEALTH DATA HUB**, *Co-PI*, "Management of Patients with Breast Cancer", 250K CAD

2023 **Amazon Research Awards**, *The awards program offers unrestricted funds and AWS Promotional Credits to support research at academic institutions and non-profit organizations in areas of AI*, (website), 60K USD

2022 – 2023 **Bourse de recherche Mitacs Globalink - Centre National de la Recherche Scientifique (CNRS)**, *PI*, "Making AI Ready for Safety-Critical Applications", 24 K CAD

2021 – 2022 **Collaborative industrial project - 360Learning - (industry collaboration)**, *Co-PI*, "Collaborative Learning Score", 160 K Euros

2020 – 2023 **PSPC Aida (BPI France) - Université Paris Saclay - IBM France -**, *Co-PI*, Safety AI, 240 K Euros

2019 – 2022 **Grant DATAIA - Institut de convergence**, *Co-PI*, Leakage of Sensitive Training Data from Deep Neural Networks, 220 K Euros

2018 – 2020 **H2020-MSCA-IF-2017 - Marie Sklodowska-Curie Global Fellowships - European Commission**, *PI*, Information Theory beyond Communications: Distributed Representations and Deep Learning, 180 K Euros

Doctoral Supervision and Research Mentoring

My mentoring activity spans doctoral supervision, co-tutelle arrangements, postdoctoral training, and research guidance across France, Canada, US, Austria, Argentina, Chile, and international collaborations. It reflects a sustained commitment to graduate education, interdisciplinary AI research, several industry collaborations, and the development of early-career researchers.

12 Current Ph.D. students (3 graduating in 2026):

1. **Mr. Guillermo Martin**, *Ph.D.* ***(co-tutelle)*** *student at Ecole de Technologie Supérieure (ÉTS) and Université Paris-Saclay, affiliated with LMF, ILLS, and Mila - Quebec AI Institute*, topic: "Belief Shaping and Strategic Communication in Multi-Agent AI Systems", Started in Winter 2026
   Co-advised at 33% with Prof. Christian Desrosiers (ILLS, ÉTS) and Prof. Benedikt Bollig (LMF, Université Paris-Saclay).

2. **Mr. Thomas Hennequin**, *Ph.D. student at Ecole de Technologie Supérieure (ÉTS), collaboration with Safran Canada*, topic: "AI-based Control Method for Aircraft Flight Assistance", Started in Autumn 2026
   Co-advised at 50% with Philippe Feyel (Safran).

3. **Ms. Cléa Chataigner**, *Ph.D **(co-tutelle)** student at McGill University and Université Paris-Saclay, affiliated with Mila - Quebec AI Institute and ILLS*, topic: "Human-In-The-Loop Auditing for Generative AI Systems", Started in Fall 2025
Co-advised at 50% with Prof. Golnoosh Farnadi (Mila, McGill University | Google).

4. **Mr. Antoine Andurao**, *Ph.D. student at Université Paris-Saclay, CIFRE with GE Healthcare, affiliated with MICS and ILLS*, topic: "Foundation Models for 3D Medical Imaging", Started in Winter 2025
Co-advised at 50% with Prof. Maria Vakalopoulou (MICS, CentraleSupélec).

5. **Mr. Matteo Sammut**, *Ph.D. **(co-tutelle)** student at Ecole de Technologie Supérieure (ÉTS) and Paris Dauphine University, affiliated with LAMSADE, ILLS, and Mila - Quebec AI Institute*, topic: "Toward a Theoretical Framework for Auditing Generative AI Models", Started in Fall 2025
Co-advised at 33% with Prof. Yan Chevaleyre (LAMSADE, Paris-Dauphine University) and Prof. Rafael Pinot (LPSM, Sorbonne Université).

6. **Ms. Chiara Roverato**, *Ph.D. student at McGill University, affiliated with Mila - Quebec AI Institute and ILLS*, topic: "Diffusion Models for Tabular Data", Started in Winter 2025
Co-advised at 50% with Prof. Mark Coates (ILLS, Mila, McGill University).

7. **Mr. Andrew McDonald**, *Ph.D. student at Université Paris-Saclay, CIFRE with Ekimetrics, affiliated with ILLS*, topic: "Building Foundation Models for Tabular Data", Started in Fall 2025
Co-advised at 50% with Dr. Spyros Angelopoulos (ILLS, CNRS).

8. **Mr. Lautaro Estienne**, *Ph.D. student at Université Paris-Saclay, affiliated with LISN and ILLS*, topic: "Smart Conversations: Transforming Human-Machine Interaction", Started in Winter 2025
Co-advised at 50% with Prof. Nona Naderi (LISN, Université Paris-Saclay)

9. **Ms. Zahra Dehghani Tafti**, *Ph.D. student at Ecole de Technologie Supérieure (ÉTS), affiliated with ILLS*, topic: "Machine Unlearning", Started in Autumn 2023
Co-advised at 50% with Prof. Mohammad Hadi Shateri (LIVIA, ÉTS Montréal).

10. **Mr. Arthur Satouf**, *Ph.D. student at Paris-Saclay University, CIFRE with Air Liquide*, Topic: "Information Retrieval", Started in 2024
Principal advisor and co-advising at 33% with Dr. Habiboulaye Amadou-Boubacar (Air Liquide) and Benjamin Piwowarski (CNRS, ISIR).

11. **Mr. Matthieu Dubois**, *Ph.D .student at Sorbonne Université, affiliated with ISIR*, topic: "Detecting AI Generated Text", Started in Fall 2023
Co-advised at 50% with Dr. François Yvon (ISIR, CNRS).

12. **Mr. Philippe Formont**, *Ph.D **(co-tutelle)** student at Université Paris-Saclay and Ecole de Technologie Supérieure (ETS)*, Topic: "Computational Tools for Molecule Discovery", Started in Spring 2023
Principal advisor and co-advising at 50% with Prof. Ismail Ben Ayed (ETS, LIVIA - ILLS).

22 Former Ph.D. students:

1. **Dr. Eric Aubinais (Fellow researcher at KTH)**, *Ph.D from Paris-Saclay University*, Topic: "Conformal Learning with Applications to Machine Learning", Graduated in 2025
   Co-advising at 50% with Prof. Elisabeth Gassiat (LMO, Université Paris-Saclay)

2. **Dr. Maxime Darrin (Mistral AI, France)**, *Ph.D (co-tutelle) from Paris-Saclay University and McGill University*, Topic: "Metrics and Methods for Generative AI Model Evaluation", Graduated in 2025
   Principal advisor and co-advising at 50% with Prof. Jackie Cheung (McGill University, Canada).

3. **Dr. Eduardo DADALTO CAMARA GOMES (Helsing France)**, *Ph.D from Paris-Saclay University*, Topic: "Securing the use of AI", Graduated in 2024
   Principal advisor with co-advising at 70% with Dr. Florence Alberge (L2S, Université Paris-Saclay).

4. **Dr. Ganesh Del Grosso (Researcher/Developer in AI and Cybersecurity at Situation.sh, France)**, *Ph.D from École Polytechnique*, Topic: "Leakage of Sensitive Training Data from Deep Neural Networks", Graduated in 2023
   Co-advising at 50% with Dr. Catuscia Palamidessi (LIX, Inria).

5. **Dr. Sebastian Espinosa (Engineering Researcher, Chile)**, *Ph.D from Universidad de Chile*, Topic: "Learning Invariances in Deep Learning", Graduated in 2023
   Co-advising at 50% with Prof. Jorge Silva (University of Chile, Chile).

6. **Dr. Marine Picot (Research Scientist at Digeiz, France)**, *Ph.D (co-tutelle) from Université Paris-Saclay and McGill University*, Topic: "Artificial Intelligence in Smart Grids", Graduated in 2023 from Paris-Saclay University and McGill University
   Principal advisor with co-advising at 50% with Prof. F. Labeau (McGill University, Canada) and collaborating with Prof. Francisco Messina (Universidad de Buenos Aires, Argentina).

7. **Dr. Malik Boudiaf (Co-Founder - Equall, Canada)**, *Ph.D from Ecole de Technologie Supérieure (ETS)*, Topic: "Information measures for low-shot learning", Graduated in 2023
   Co-advising at 50% with Prof. Ismail Ben Ayed (ÉTS, Canada).

8. **Dr. Federica Granese (Junior Researcher at INRIA)**, *Ph.D (co-tutelle) from Institut Polytechnique de Paris and Università degli Studi di Roma*, Topic: "Information-Theoretic Methods for Safety AI", Graduated in 2023
   Co-advising at 25% with Dr. Catuscia Palamidessi (LIX, Inria, France) and Daniele Gorla (Università degli Studi di Roma, Italy).

9. **Dr. Mohammad Hadi Shateri (Assistant Professor at ÉTS Montreal)**, *Ph.D from McGill University*, Topic: "Deep Learning and Privacy in Smart Grids", Graduated in 2022
   Co-advising at 50% with Prof. F. Labeau (McGill University, Canada) and Prof. Francisco Messina (Universidad de Buenos Aires, Argentina).

10. **Dr. Matias Vera (Research Scientist at Conicet)**, *Ph.D from Universidad de Buenos Aires (Argentina)*, Topic: "Learning via Information-Bottleneck Methods", Graduated in 2020
Co-advised at 50% with Prof. Leonardo Rey Vega (UBA - CONICET, Argentina).

11. **Dr. Clément Feutry (Project Manager - Groupe BPCE)**, *Ph.D from Université Paris-Saclay*, Topic: "Two Sides of Relevant Information: Anonymized Representation through Deep Learning and Predictor Monitoring", Graduated in 2019
Co-advising at 70% with Dr. P. Duhamel (CNRS).

12. **Dr. Kai Wan (Assistant Professor at Huazhong University of Science and Technology)**, *Ph.D from Université Paris-Saclay*, Topic: "Fundamental Limits of Index Coding", Graduated in 2018
Principal advisor with co-advising at 50% with Prof. D. Tuninetti (University of Chicago, US).

13. **Dr. Georg Pichler (Assistant Professor at TU Wien)**, *Ph.D from TU Wien (Austria), Topic: "Information-Theoretic Clustering"*, TUWien, Austria, Graduated in 2017
Co-advised at 50% with Prof. G. Matz (TUWien, Austria).

14. **Dr. Gil Katz (Co-founder | Innovation enthusiast | Mentor @TenGiving)**, *Ph.D from Université Paris-Saclay (France)*, Topic: "Distributed Statistical Inference with Communication Constraints", Graduated in 2017
Co-advising at 50% with Prof. M. Debbah (CentraleSupélec).

15. **Dr. Maggie Mhanna (AI Field Solutions Architect at Google)**, *Ph.D from Université Paris-Saclay* , Topic: "Distributed Hypothesis Testing with Privacy Constraints", Graduated in 2017
Co-advising at 50% with Dr. P. Duhamel (CNRS).

16. **Dr. Chao He (Researcher at Ericsson in Stockholm)**, *Ph.D from Université Paris-Saclay*, Topic: "Cooperation and Feedback in Wireless Networks", Graduated in 2016
Co-advising at 30% with Prof. S.Yang (CentraleSupélec).

17. **Dr. Meryem Benammar (Associate Professor at ISAE SUPAERO)**, *Ph.D from Supélec*, Topic: "Broadcast Channels: Side Information and Security Constraints", Graduated in 2016
Principal advisor.

18. **Dr. German Bassi (Researcher at Ericsson Research)**, *Ph.D from Supélec*, Topic: "Interference Relay Networks: Feedback and Security Constraints", Graduated in 2015
Co-advised at 50% with Prof. S. Yang (CentraleSupélec).

19. **Dr. Andres Altieri (Staff engineer at Qualcomm)**, *Ph.D (co-tutelle) from Supélec and Universidad de Buenos Aires (Argentina)*, Topic:"Fundamental Limits of Cooperative Large Wireless Networks", Graduated in 2014
Co-advised at 50% with Prof. C. Galarza (UBA - CONICET, Argentina) and Prof. Leonardo Rey Vega (UBA - CONICET, Argentina).

20. **Dr. Joffrey Villard (CTO at Syntelier)**, *Ph.D from Supélec*, Topic: "Task-Oriented Source Coding: Secure Transmission, Detection", Graduated in 2012
Co-advised at 50% with Pascal Bianchi (Télécom ParisTech).

21. **Dr. Arash Behboodi (Director of Engineering at Qualcomm AI Research)**, *Ph.D from Supélec*, Topic: "Cooperative Networks with Channel Uncertainty", Graduated in 2012 from Supélec
Advising at 90% (CentraleSupélec).

22. **Dr. Patricia Layec (Team lead & researcher at Nokia Bell Labs)**, *Ph.D from Supélec, CIFRE Orange*, Topic: "Information-Theoretic Bounds and Practical Coding Schemes for MIMO Communications Under Imperfect CSIR and Partial and/or Imperfect CSIT", Graduated in 2009
Co-advising at 20% with Prof. Antoine O. Berthet (CentraleSupélec).

List of Postdoctoral Fellows:

1. **Dr. Andres Altieri (Staff engineer at Qualcomm)**, *"Securing the use of AI"*, June 2023 – May 2024, (L2S, Paris-Saclay University).

2. **Dr. Pierre Colombo (VLM & Multimodal Research @ Cohere)**, *"Securing the use of AI"*, since January 2022 – September 2022, (L2S, Paris-Saclay University).

3. **Dr. Marco Romanelli (Assistant Professor at Hofstra University)**, *"Securing the use of AI"*, since Nov. 2020 – September 2022, (L2S, Paris-Saclay University).

4. **Dr. Hamid Jalalzai**, *"Bounding Information Leakage in Machine Learning"*, February 2021 – September 2022, In collaboration with Dr. Catuscia Palamidessi (LIX, Inria, France).

5. **Dr. Matias Vera (Researcher at CONICET)**, *"Generalization, Information and Learning"*, April 2019 – 2021, In collaboration with Prof. Leonardo Rey Vega (UBA - CONICET, Argentina).

6. **Dr. Francisco Messina (Assistant Professor at UBA)**, *"Artificial Intelligence in Smart Grids: Robust Learning"*, June 2018 - October 2020, In collaboration with Prof. F. Labeau (McGill University, Canada).

7. **Dr. Georg Pichler (Assistant Professor at TU Wien)**, *"Estimation of Information Measures based on Neural Networks"*, July 2018 - Sep. 2019, (L2S, Paris-Saclay University).

8. **Zuxing Li (Assistant Professor at Tongji University)**, *"Distributed Compression and Learning on Power Lines"*, March 2018 - February 2020, In collaboration with RTE France and Prof. M. Kieffer (Paris-Saclay University)
(L2S, Université Paris Saclay).

Selected Honors and Distinctions

2026 - 2030 **Holder of the AI Cluster DATAIA Paris-Saclay - Chaires modulaires**, *Mathematical Principles of Learning and Communication in Attention-Based Models*, Jointly with Prof. Claire Boyer at Laboratoire des Mathématiques d'Orsay (LMO, UPSaclay), 500K Euros

2024 **Area Chair Award**, *Association for Computational Linguistics (ACL 2024)*

2023 **Amazon Research Awards**, *The awards program offers unrestricted funds and AWS Promotional Credits to support research at academic institutions and non-profit organizations in areas of AI*, (website)

2022 **AAAI Outstanding Student Paper Award**, InfoLM: A New Metric to Evaluate Summarization & Data2Text Generation, AAAI Conference on Artificial Intelligence

Jan. 2018 **H2020-MSCA-IF-2017 - Marie Sklodowska-Curie Global Fellowship**, *Information Theory beyond Communications: Distributed Representations and Deep Learning*, European Commission (7% of acceptance rate), 2 years

2016 **IEEE Senior Member**

April 2016 **Visiting Professor Fellowship**, *Information Processing Group, Ecole Polytechnique Federale Lausanne (EPFL)*, Lausanne, Switzerland, 1 week

2014 **Coauthor of Best Paper Award**, *Young Researcher Award from Newcom#*

2003 – 2005 **Saint-Exupéry Fellowship**, *French Embassy in Argentina*, France

2001 – 2002 **Merit Award Scholarship**, *University of Buenos Aires*, Buenos Aires, Argentina

Publication Record

- **60 Publications in journals:**
  - **Applied Mathematics:** IEEE Trans. on Information Theory, The Annals of Applied Probability, Information and Inference: A Journal of the IMA.
  - **Computer Science:** Machine Learning, Journal of Machine Learning Research (JMLR), IEEE Trans. on Pattern Analysis and Machine Intelligence, Transactions on Machine Learning Research (TMLR), Neurocomputing.
  - **Communication, Signal and Smart Grids:** IEEE Trans. on SP, IEEE Trans. on Signal and Information Processing over Networks, IEEE Trans. on Com., IEEE Trans. on Wireless Com., IEEE/ACM Trans. on Networking, Entropy, IEEE JSAC, IEEE Trans. on Smart Grids, IEEE Internet of Things Journal.
  - **Security and Privacy:** IEEE Trans. on Inf. Forensics and Security.
- **189 Publications in proceedings of international conferences:**
  - **Computer Science:** NeurIPS, ICML, ICLR, AISTATS, CCS, ECCV, CVPR, ACL, EMNLP, AAAI, Eurospeech,...
  - **Information Theory:** IEEE ISIT, IEEE ITW.
  - **Communication, Signal and Smart Grids:** IEEE ICASSP, IEEE ICC, IEEE Globecom,...
  - **+20** invited papers to special sessions.
- **+50 Invited lectures** (universities, tutorials and workshops).

International Scientific Leadership and Service

Coordination of scientific activities (selected subset)

Feb 18 - 20, 2026 **General Chair**, *Workshop: AI for Health*, ETS Montreal, (website)
(up to 110 participants on site)

Oct 1 - 3, 2024 **General Chair**, *Workshop: NLP in the era of generative AI, cognitive sciences, and societal transformation*, Mila - Quebec AI Institute, (website)
(up to 250 participants on site and 3000 online)

May 24 - 26, 2023 **General Co-Chair**, *Workshop ILLS - DATAIA*, Paris-Saclay University, (website)

May 18, 2023 **General Co-Chair**, *Séminaire des IRL CRM et ILLS*, Pôle CRM de l'UQAM, (website)

2021 - 2023 **Conference Committee Chair**, *IEEE Information Theory Society*

July 2019 **General Co-Chair of the 2019**, *IEEE International Symposium on Information Theory (ISIT) in Paris*, IEEE Information Theory Society, (the most prestigious conference in the field with around 1100 participants)
`https://2019.ieee-isit.org`

Nov. 2016 **Conference Special Session**, *Information Theory and Statistical Learning*, Asilomar Conference on Signals, Systems, and Computers

Outreach and participation in scientific committees (selected subset)

2024 - present **Area Chair and Senior Area Chair**, *ICML 2026, NeurIPS 2026, ARR 2026, AAAI 2025-2027, CCS 2025, USENIX 2025, ISIT, and ITW*

2022 - 2023 **International expert for committee evaluation program TRAIL**, *Trusted AI Labs*, Walloon and Brussels regions

2021 - 2022 **Vice-president of the expert committee 23 (Artificial Intelligence) of the ANR for the evaluation of call for research projects**, *ANR*, National Research Agency

2020 - 2021 **Member of the expert committee 23 (Artificial Intelligence) of the ANR for the evaluation of call for research projects**, *ANR*, National Research Agency

April 2018 - 2021 **Member of the evaluation committee of ERC Starting Grants**, *European Research Council*, European Commission

Nov. 2020 - 2024 **Board of Entropy**, *Topic Editor of Information Theory, Probability and Statistics*

2019 - 2021 **Associate Editor**, *IEEE Transactions on Information Forensics and Security*

2010 - Present **Session Chair**, *IEEE Intern. Symposium on Information Theory (ISIT) and IEEE Workshop on Information Theory (ITW)*

Service on Ph.D. committees (selected subset)

Dec. 3, 2024 **Ph.D. examiner/president of jury**, *Université Paris-Saclay*, by Alexandre Perez-Label, Évaluation de classifieurs probabilistes et impact sur la décision

Dec. 11th, 2023 **Ph.D. examiner**, *Université Paris-Saclay*, by Fabio Arnez, Deep Neural Network Uncertainty Runtime Monitoring for Robust and Safe AI-based Automated Navigation

Dec. 13th, 2021 **Ph.D. examiner/president of jury**, *Université Paris-Saclay*, by Victor Bouvier, Vers l'apprentissage adaptatif à l'aide de représentations invariantes

Nov. 17th, 2021 **Ph.D. examiner**, *Institut Polytechnique de Paris, Télécom ParisTech*, by Pierre Colombo, Apprendre à représenter et à générer du texte en utilisant des mesures d'information

Nov. 6th, 2020 **Ph.D. reviewer**, *Institut Polytechnique de Paris, Télécom ParisTech*, by Hamid Jalalzai, Learning from Multivariate Extremes: Theory and Application to Natural Language Processing

International and French Collaborations

- At the international level, collaborations have been established with TU Wien (Austria), TU Berlin (Germany), UCLouvain (Belgium), Technion (Israel), McGill (Canada), University of Montreal (Canada), ETS (Canada), INRS (Canada), Universidad de Chile (Chile), Universidad de Buenos Aires (Argentina), University of Chicago (USA), New York University (USA), University of Michigan (USA), among others.
- At the national level, collaborations have been established with Telecom ParisTech, INRIA (Lix - Ecole Polytechnique), Sorbonne université (ISIR), ENS Lyon (LIP), Université Aix-Marseille (LIS), University Paris-Saclay (LMO, LISN), Université de Cergy (ENSEA), among others.

Teaching, Graduate Training and Curriculum Leadership

Courses at Master (M2) Level (selected subset)

2019 – 2020 **Foundations of Privacy and Machine Learning**, *Parisian Master of Research in Computer Science (MPRI)*, ENS Ulm, Ecole Polytechnique, University Paris Diderot, Telecom Paris Tech, University Paris Diderot
(4 lectures of 3h each per year)

2018 – 2021 **Introduction to Deep Learning**, *International summer school on Artificial Intelligence*, Direction International of CentraleSupélec, CentraleSupélec
(2 lectures of 2h30 each per year)

2017 – 2018 **Deep Learning and IoT Data Analytics**, *Master Advanced Wireless Communications Systems (SAR)*, Research master M2, CentraleSupélec
(4 lectures of 3h each per year)

2015 – 2018 **Mathematics of Information and Source Coding**, *Master Multimedia Networking (MN)*, Master M2, Telecom ParisTech/CentraleSupélec/Université Paris-Sud
(8 lectures of 3h each per year)

2009 – 2018 **Elements of Information Theory**, *Master Advanced Wireless Communications Systems (SAR)*, Research master M2, CentraleSupélec
(5 lectures of 3h each per year)

2009 – 2016 **Network Information Theory**, *Master Advanced Wireless Communications Systems (SAR)*, Research Master M2, CentraleSupélec
(4 lectures of 3h each per year)

Courses/TAs/TPs at Engineering Level (selected subset)

2021 – present **Deep Learning**, 3rd year of engineering program at CentraleSupélec
(24h of lectures per year)

2020 **Research Projects**, 2nd and 3rd year of engineering program at CentraleSupélec
(24h per year)

2020 – 2021 **Information Theory, Large Dimensional Data and AI**, 2nd year (SG8) of engineering program at CentraleSupélec
(2 lectures of 3h each per year)

2017 – 2018 **Information Theory and Information Complexity**, *1st year of engineering program at CentraleSupélec*, ST4 - responsible of the course and curriculum design (replaced by R. Combes during sabbatical leave)
(around 18h per year)

Around 200h per year, according to the French system (equivalent TD), until Dec. 2021.

Education and Degrees

May 2019 **Habilitation à Diriger les Recherches (HDR)**, *Université Paris Saclay (May 2019)*, France, (ultimate French university degree to serve as Ph.D supervisor)
Information-Theoretic Methods: From Communication to Decision, Privacy and Learning, at the Heart of Data Sciences

May 2007 **Ph.D in Physics**, *Université Paris Sud*, Orsay, France
Multi-user information theory: State information and imperfect channel knowledge

Dec. 2003 **MSc/BSc Mathematics and Electrical Engineering**, *Faculty of Engineering and Mathematics, University of Buenos Aires*, Buenos Aires, Argentina

Dec. 2000 **BSc Electrical Engineering**, *(with Honors) Faculty of Engineering, University of Buenos Aires*, Buenos Aires, Argentina

Earlier Research Experience

Feb. 2004 – Sep. 2004 **Visting Researcher**, *Institut für Nachrichtentechnik und Hochfrequenztechnik*, Vienna, Austria, (working on Capacity of Time-Varying channels)

July 2002 – Dec. 2002 **Master Internship**, *Stanford Research Institute (SRI)*, California, USA, (working on Statistical Language Models)

March 2001 – July 2002 **Young Research Associate**, *Institute of Biomedical Engineering, Faculty of Engineering*, Buenos Aires, Argentina, (working on Speech Recognition)

Invited Seminars, Tutorials and Outreach (subset)

1. Rational Speech Acts as KL Projections: From Pragmatic Reasoning to Multi-Agent Communication, IVADO Regroupement 2 Workshop Day, Montreal, August 2026.
2. Generative AI in the Age of Multi-Agent Reasoning: *Evaluation, coordination, and scientific discovery*, COMMEX annual meeting of Air Liquide - Keynote IA, France, April 13th, 2026.
3. Generative AI and Scientific Discovery, R7+ Meeting - Keynote IA, Ottawa, November 2025.
4. Information-Theoretic Methods for Trustworthy Machine Learning, STARaCom Annual Meeting at McGill University, Montréal, QC, June 6th, 2023.
5. AI and Data Privacy, Keynote at the Symposium on Artificial Intelligence | MIT-France, July 12th to 13th, 2022: (online)
6. Information-Theoretic Methods in Data Sciences: Model Uncertainty, Robustness and Model Drift, Keynote at Journée Statistique et Informatique pour la Science des Données à Paris Saclay, January 26th, 2022: (online)
7. Information Theory for / by Deep Learning, Keynote at Research Days of Labex Digicosme, November 9th, Université Paris-Saclay (virtual talk), 2020: (online)
8. Learning Anonymized Representations with Mutual Information, MLAI 2019 – International Workshop on Machine Learning and Artificial Intelligence, Paris, October 7th-8th, 2019: (online)
9. Les clés de l'IA: le traitement et l'optimisation de l'information, more than 450 participants (undergrads, Ph.D. students, industry, interested public) at ÉTS, Montréal, Canada, October 16th, 2018. Part I: (online) and Part II: (online)
10. Information-Theoretic Biclustering and Distributed Binary Detection, Royal Institute of Technology in Stockholm (KTH), Stockholm, Sweden, Sep. 14th, 2017.
11. Exploring Connections between Information Theory, Machine Learning and Statistics, Institut National de la Recherche Scientifique (INRS), Montreal, Canada, September 1st, 2016.
12. The Compound Broadcast Channel: Recent Advances and Challenges, Information Processing Group, Ecole Polytechnique Federale Lausanne (EPFL), Switzerland, July 23rd, 2015.
13. On the Multiterminal Interactive Lossy Source Coding Problem, Dept. of EE, Imperial College London, UK, September 17th, 2014.
14. A Proof of the Generalized Markov Lemma with Countable Infinite Alphabets, Dept. of Mathematics, University of Hong Kong, Hong Kong, June 3rd, 2014.
15. An Information-Theoretic Approach to Secure Transmission and Storage of Data Sources, Technische Universität Wien (TUW), Vienna, Austria, April 4th, 2014.
16. Capacity of Simultaneous State-Dependent Channels with State Information at the Transmitter, EE Colloquium, Israel Institute of Technology, Technion, Haifa, Israel, April 23, 2009.

Full Publication List

Preprints and submitted/revised articles

[P1] P. Formont, M. Darrin, I. B. Ayed, and P. Piantanida, "MolRGen: A Training and Evaluation Setting for De Novo Molecular Generation with Reasonning Models," 2026. [paper] [arXiv]

[P2] P. Piantanida and G. Pichler, "A theory for rational speech acts as KL projections with powered semantics," 2026, (submitted)

[P3] Y. Bahram, Z. Dehghani, M. Desbos, E. Granger, P. Piantanida, and M. Shateri, "Continuous adversarial meanflow transfer," 2026. [paper] [arXiv]

[P4] C. Chataigner, P. Piantanida, and G. Farnadi, "Position: Auditing Is Not Evaluating; LLM Audit Requires Dynamic, Contextual, Budget-Aware and Reliable Evidence," Feb. 2026, working paper or preprint. [paper]

[P5] L. Estienne, E. Ernst, M. Vera, P. Piantanida, and L. Ferrer, "Ecuas$_n$: A family of metrics for principled evaluation of uncertainty-augmented systems," 2026. [paper] [arXiv]

Proceedings of international conferences

[C1] A. Satouf, G. D'Erasmo, Y. Zong, H. Amadou-Boubacar, P. Piantanida, and B. Piwowarski, "STORM: Stepwise token optimization with reward-guided beam search," in *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing*, 2026. [paper] [arXiv]

[C2] L. Fosse, P. Formont, E. Aubinais, and P. Piantanida, "Data-driven estimation of statistical deficiency," in *2026 IEEE Information Theory Workshop (ITW)*, 2026

[C3] E. Aubinais, P. Formont, P. Piantanida, and E. Gassiat, "Membership Inference Risks in Quantized Models: A Theoretical and Empirical Study," in *The 29th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2026. [paper] [arXiv]

[C4] G. Pichler, M. Romanelli, and P. Piantanida, "Happiness as a Measure of Fairness," in *The 29th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2026. [paper] [arXiv]

[C5] A. Satouf, Y. Zong, H. A. Boubacar, P. Piantanida, and B. Piwowarski, "QueStER: Query Specification for Generative Keyword-Based Retrieval," in *Findings of the Association for Computational Linguistics: EACL 2026*, ser. Findings of ACL, V. Demberg, K. Inui, and L. Marquez, Eds. Association for Computational Linguistics, March 2026, pp. 5957–5968. [paper] [arXiv]

[C6] A. Satouf, G. B. Zenou, B. Piwowarski, H. A. Boubacar, and P. Piantanida, "Rational Retrieval Acts: Leveraging Pragmatic Reasoning to Improve Sparse Retrieval," in *Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval*, N. Ferro, M. Maistro, G. Pasi, O. Alonso, A. Trotman, and S. Verberne, Eds. ACM, 2025, pp. 2674–2678. [paper] [arXiv]

[C7] P. Marza, L. Fillioux, S. Boutaj, K. Mahatha, C. Desrosiers, P. Piantanida, J. Dolz, S. Christodoulidis, and M. Vakalopoulou, "THUNDER: Tile-level Histopathology image UNDERstanding benchmark," in *Advances in Neural Information Processing Systems 38*. Neural Information Processing Systems Foundation, Inc. (NeurIPS), 2025, pp. 171 942–172 034, (spotlight). [paper] [arXiv]

[C8] P. Formont, M. DARRIN, B. Karimian, E. Granger, J. C. Cheung, I. B. Ayed, M. Shateri, and P. Piantanida, "Learning Task-Agnostic Representations through Multi-Teacher Distillation," in *Advances in Neural Information Processing Systems 38*. Neural Information Processing Systems Foundation, Inc. (NeurIPS), 2025, pp. 121 561–121 607. [paper] [arXiv]

[C9] C. Spinoso-Di Piano, D. E. Austin, P. Piantanida, and J. C. Cheung, "$(RSA)^2$: A Rhetorical-Strategy-Aware Rational Speech Act Framework for Figurative Language Understanding," in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, W. Che, J. Nabende, E. Shutova, and M. T. Pilehvar, Eds. Association for Computational Linguistics, July 2025, pp. 20 898–20 938. [paper] [arXiv]

[C10] L. Estienne, G. B. Zenou, N. Naderi, J. C. Cheung, and P. Piantanida, "Collaborative Rational Speech Act: Pragmatic Reasoning for Multi-Turn Dialog," in *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, November 2025, pp. 22 509–22 523, (oral). [paper] [arXiv]

[C11] L. Fosse, F. Bechet, B. Favre, G. Damnati, G. Lecorvé, M. Darrin, P. Formont, and P. Piantanida, "Statistical Deficiency for Task Inclusion Estimation," in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, W. Che, J. Nabende, E. Shutova, and M. T. Pilehvar, Eds. Association for Computational Linguistics, July 2025, pp. 382–415. [paper] [arXiv]

[C12] M. Dubois, F. Yvon, and P. Piantanida, "MOSAIC: Multiple Observers Spotting AI Content," in *Findings of the Association for Computational Linguistics: ACL 2025*, W. Che, J. Nabende, E. Shutova, and M. T. Pilehvar, Eds. Association for Computational Linguistics, July 2025, pp. 24 230–24 247. [paper] [arXiv]

[C13] ——, "MOSAIC at GENAI Detection Task 3 : Zero-Shot Detection Using an Ensemble of Models," in *Proceedings of the 1stWorkshop on GenAI Content Detection (GenAIDetect)*. International Conference on Computational Linguistics, January 2025, pp. 371–376. [paper]

[C14] O. Beaumont, R. Bourgouin, M. Darrin, L. Marchal, and P. Piantanida, "Leveraging Expert Usage to Speed up LLM Inference with Expert Parallelism," in *Euro-Par 2025: Parallel Processing - 31st European Conference on Parallel and Distributed Processing, Dresden, Germany, August 25-29, 2025, Proceedings, Part I*, ser. Lecture Notes in Computer Science, W. E. Nagel, D. Goehringer, and P. C. Diniz, Eds., vol. 15900. Springer, 2025, pp. 145–158. [paper]

[C15] M. Dubois, F. Yvon, and P. Piantanida, "How Sampling Affects the Detectability of Machine-written texts: A Comprehensive Study," in *Findings of the Association for Computational Linguistics: EMNLP 2025*, C. Christodoulopoulos, T. Chakraborty, C. Rose, and V. Peng, Eds. Suzhou, China: Association for Computational Linguistics, November 2025, pp. 11 369–11 387. [paper] [arXiv]

[C16] C. Presvôts, M. Kieffer, T. Prevost, P. Panciatici, Z. Li, and P. Piantanida, "Two-stage Multiple-Model Compression Approach for Sampled Electrical Signals," in *2024 Data Compression Conference (DCC)*, A. Bilgin, J. E. Fowler, J. Serra-Sagristà, Y. Ye, and J. A. Storer, Eds. IEEE, 2024, pp. 522–531. [paper]

[C17] M. Darrin, P. Formont, I. B. Ayed, J. C. Cheung, and P. Piantanida, "When is an Embedding Model More Promising than Another?" in *Advances in Neural Information Processing Systems 37*. Neural Information Processing Systems Foundation, Inc. (NeurIPS), 2024, pp. 68 330–68 379. [paper] [arXiv]

[C18] A. Altieri, M. Romanelli, G. Pichler, F. Alberge, and P. Piantanida, "Beyond the Norms: Detecting Prediction Errors in Regression Models," in *Proceedings of the 41st International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 235. PMLR, 21–27 Jul 2024, pp. 1186–1221, (spotlight). [paper] [arXiv]

[C19] E. D. C. Gomes, M. Romanelli, G. Pichler, and P. Piantanida, "A Data-Driven Measure of Relative Uncertainty for Misclassification Detection," in *The Twelfth International Conference on Learning Representations*, 2024. [paper] [arXiv]

[C20] M. Darrin, P. Formont, J. Cheung, and P. Piantanida, "COSMIC: Mutual Information for Task-Agnostic Summarization Evaluation," in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, August 2024, pp. 12 696–12 717, (oral). [paper] [arXiv]

[C21] M. Darrin, I. Arous, P. Piantanida, and J. Cheung, "GLIMPSE: Pragmatically Informative Multi-Document Summarization for Scholarly Reviews," in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, August 2024, pp. 12 737–12 752. [paper] [arXiv]

[C22] F. Granese, M. Romanelli, and P. Piantanida, "Optimal Zero-Shot Detector for Multi-Armed Attacks," in *Proceedings of The 27th International Conference on Artificial Intelligence and Statistics*, ser. Proceedings of Machine Learning Research, S. Dasgupta, S. Mandt, and Y. Li, Eds., vol. 238. PMLR, 02–04 May 2024, pp. 2467–2475. [paper] [arXiv]

[C23] D. Gorla, L. Jalouzot, F. Granese, C. Palamidessi, and P. Piantanida, "On the (Im)Possibility of Estimating Various Notions of Differential Privacy (short paper)," in *Proceedings of the 24th Italian Conference on Theoretical Computer Science, Palermo, Italy, September 13-15, 2023*, ser. CEUR Workshop Proceedings, G. Castiglione and M. Sciortino, Eds., vol. 3587. CEUR-WS.org, 2023, pp. 219–224. [paper] [arXiv]

[C24] P. Colombo, M. Picot, N. Noiry, G. Staerman, and P. Piantanida, "Toward Stronger Textual Attack Detectors," in *Findings of the Association for Computational Linguistics: EMNLP 2023*, H. Bouamor, J. Pino, and K. Bali, Eds. Association for Computational Linguistics, 2023, pp. 484–505. [paper] [arXiv]

[C25] M. Darrin, G. Staerman, E. D. C. Gomes, J. C. Cheung, P. Piantanida, and P. Colombo, "Unsupervised Layer-wise Score Aggregation for Textual OOD Detection," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38. Association for the Advancement of Artificial Intelligence (AAAI), February 2024, pp. 17 880–17 888. [paper] [arXiv]

[C26] P. Colombo, M. Peyrard, N. Noiry, R. West, and P. Piantanida, "The Glass Ceiling of Automatic Evaluation in Natural Language Generation," in *Findings of the Association for Computational Linguistics: IJCNLP-AACL 2023 (Findings)*, J. C. Park, Y. Arase, B. Hu, W. Lu, D. Wijaya, A. Purwarianti, and A. A. Krisnadhi, Eds. Nusa Dua, Bali: Association for Computational Linguistics, November 2023, pp. 178–183. [paper] [arXiv]

[C27] M. Darrin, P. Piantanida, and P. Colombo, "RainProof: An Umbrella to Shield Text Generator from Out-Of-Distribution Data," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, H. Bouamor, J. Pino, and K. Bali, Eds. Singapore: Association for Computational Linguistics, December 2023, pp. 5831–5857. [paper] [arXiv]

[C28] P. Colombo, N. Noiry, G. Staerman, and P. Piantanida, "A Novel Information Theoretic Objective to Disentangle Representations for Fair Classification," in *Findings of the Association for Computational Linguistics: IJCNLP-AACL 2023 (Findings)*, J. C. Park, Y. Arase, B. Hu, W. Lu, D. Wijaya, A. Purwarianti, and A. A. Krisnadhi, Eds. Nusa Dua, Bali: Association for Computational Linguistics, November 2023, pp. 184–198. [paper] [arXiv]

[C29] P. Colombo, V. Pellegrain, M. Boudiaf, M. Tami, V. Storchan, I. Ayed, and P. Piantanida, "Transductive Learning for Textual Few-Shot Classification in API-based Embedding Models," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, H. Bouamor, J. Pino, and K. Bali, Eds. Singapore: Association for Computational Linguistics, December 2023, pp. 4214–4231. [paper] [arXiv]

[C30] M. Boudiaf, E. Bennequin, M. Tami, C. Hudelot, A. Toubhans, P. Piantanida, and I. B. Ayed, "Open-Set Likelihood Maximization for Few-Shot Learning," in *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2023, pp. 24 007–24 016. [paper] [arXiv]

[C31] P. Colombo, E. Dadalto, G. Staerman, N. Noiry, and P. Piantanida, "Beyond Mahalanobis Distance for Textual OOD Detection," in *Advances in Neural Information Processing Systems 35*, S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, and A. Oh, Eds., vol. 35. Neural Information Processing Systems Foundation, Inc. (NeurIPS), 2022, pp. 17 744–17 759. [paper] [arXiv]

[C32] N. M. Guerreiro, P. Colombo, P. Piantanida, and A. Martins, "Optimal Transport for Unsupervised Hallucination Detection in Neural Machine Translation," in *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Toronto, Canada: Association for Computational Linguistics, July 2023, pp. 13 766–13 784. [paper] [arXiv]

[C33] F. Granese, M. Picot, M. Romanelli, F. Messina, and P. Piantanida, "MEAD: A Multi-Armed Approach for Evaluation of Adversarial Examples Detectors," in *Machine Learning and Knowledge Discovery in Databases - European Conference, ECML PKDD 2022, Grenoble, France, September 19-23, 2022, Proceedings, Part III*, ser. Lecture Notes in Computer Science, M. Amini, S. Canu, A. Fischer, T. Guns, P. K. Novak, and G. Tsoumakas, Eds., vol. 13715. Springer Nature Switzerland, 2022, pp. 286–303. [paper] [arXiv]

[C34] G. Pichler, P. J. A. Colombo, M. Boudiaf, G. Koliander, and P. Piantanida, "A Differential Entropy Estimator for Training Neural Networks," in *Proceedings of the 39th International*

Conference on Machine Learning, ser. Proceedings of Machine Learning Research, vol. 162. PMLR, 17–23 Jul 2022, pp. 17 691–17 715. [paper] [arXiv]

[C35] P. Colombo, G. Staerman, N. Noiry, and P. Piantanida, "Learning Disentangled Textual Representations via Statistical Measures of Similarity," in *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Dublin, Ireland: Association for Computational Linguistics, May 2022, pp. 2614–2630, (spotlight). [paper] [arXiv]

[C36] G. Del Grosso, H. Jalalzai, G. Pichler, C. Palamidessi, and P. Piantanida, "Leveraging Adversarial Examples to Quantify Membership Information Leakage," in *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, Jun 19, 2022 - Jun 24, 2022, New Orleans, Louisiana, Nov 16 2022, pp. 10 389–10 399. [paper] [arXiv]

[C37] S. Espinosa, J. F. Silva, and P. Piantanida, "A Data-Driven Quantization Design for Distributed Testing Against Independence with Communication Constraints," in *ICASSP 2022 - 2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 22-27 May 2022 in Singapore. 2022, pp. 5238–5242. [paper]

[C38] E. Dadalto Carama Gome, F. Alberge, P. Duhamel, and P. Piantanida, "Igeood: An Information Geometry Approach to Out-of-Distribution Detection ," in *The Tenth International Conference on Learning Representations*, (Virtual) Mon Apr 25th through Fri the 29th 2022. [paper] [arXiv]

[C39] P. Colombo, C. Clavel, and P. Piantanida, "InfoLM: A New Metric to Evaluate Summarization & Data2Text Generation," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36. Association for the Advancement of Artificial Intelligence (AAAI), Feb. 22 - March 1 2022, pp. 10 554–10 562, doi:https://doi.org/10.1609/aaai.v36i10.21299 (Best Paper Award). [paper] [arXiv]

[C40] C. Pinzon, C. Palamidessi, P. Piantanida, and F. Valencia, "On the Impossibility of non-Trivial Accuracy in Presence of Fairness Constraints," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36. Association for the Advancement of Artificial Intelligence (AAAI), Feb. 22 - March 1 2022, pp. 7993–8000. [paper] [arXiv]

[C41] M. Picot, F. Messina, F. Labeau, and P. Piantanida, "Robust Autoencoder-based State Estimation in Power Systems," in *IEEE ISGT NA 2022: "Technology Solutions for an Evolving Grid"*, February 21 - 24, Washington, DC, USA 2022

[C42] G. Del Grosso, G. Pichler, C. Palamidessi, and P. Piantanida, "Formalizing Attribute and Membership Inference Attacks on Machine Learning Models," in *Privacy Preserving Machine Learning*, Nov. 19th, Virtual ACM CCS 2021 Workshop 2021. [paper]

[C43] E. Dadalto Carama Gome, F. Alberge, P. Duhamel, and P. Piantanida, "Igeood: An Information Geometry Approach to Out-of-Distribution Detection ," in *NeurIPS 2021 Workshop on Distribution Shifts*, Dec 13th, 2021, Virtual 2021

[C44] P. Colombo, G. Staerman, C. Clavel, and P. Piantanida, "Automatic Text Evaluation through the Lens of Wasserstein Barycenters," in *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, November 2021, pp. 10 450–10 466. [paper] [arXiv]

[C45] F. Granese, M. Romanelli, D. Gorla, C. Palamidessi, and P. Piantanida, "DOCTOR: A Simple Method for Detecting Misclassification Errors," in *Advances in Neural Information Processing Systems*, Dec 6th - 14th, 2021, Virtual 2021, (spotlight). [paper] [arXiv]

[C46] O. Veilleux, M. Boudiaf, P. Piantanida, and I. Ben Ayed, "Realistic evaluation of transductive few-shot learning," in *Advances in Neural Information Processing Systems*, Dec 6th - 14th, 2021, Virtual 2021. [paper] [arXiv]

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